Stromversorgung

File: power.kicad_sch

Steuerung_ESP32S3

File: ctrl_s3.kicad_sch

HMI_ESP32P4

File: hmi_p4.kicad_sch

Analogeingänge

File: analog_in.kicad_sch

Analogausgänge

File: analog_out.kicad_sch

Digital_RS485_Luefter

File: io.kicad_sch

Pumpensteuerung — AIO-Steuerplatine, Rev 0.2 (Entwurf)
Ersetzt LOGO! 8.4 + AM2 + Raspberry Pi 3B+. Zwei Controller ohne Linux:
  ESP32-S3 = Steuerung (Sensoren, PI-Druckregelung, V20 per RS485, Relais, AO, Trockenlaufschutz)
  ESP32-P4 (M5Stack Stamp-P4) = App/HMI (Web-UI, Bewässerung, Wetter, MQTT/HA, 7"-DSI-Touch, LAN)
  Verbindung: UART (S3-UART0 ↔ P4), P4 kann S3 resetten/flashen.
Versorgung 230 V AC → IRM-30-24 → 24 V → 12 V (Lüfter) / 5 V (P4, Display) → 3,3 V (S3, Logik).
Details, GPIO-Belegung und offene Punkte: hardware/SPEC.md

Erzeugt mit hardware/tools/gen_schematic.py — Änderungen bitte dort

Brunnensteuerung Wathlingen

Sheet: /
File: pumpensteuerung.kicad_sch

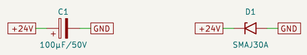
Title: Pumpensteuerung — AIO-Steuerplatine (Übersicht)

Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/7

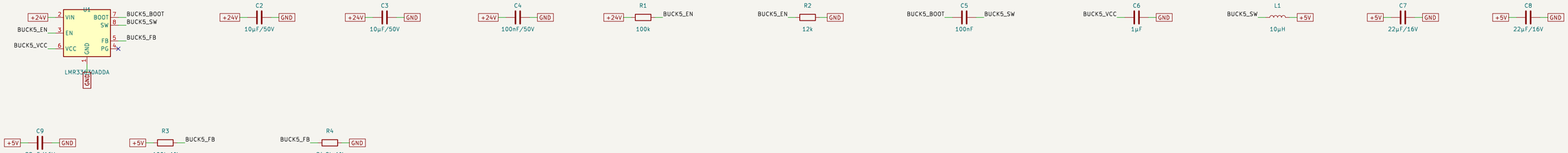
230 V AC Eingang (Netzbereich – Abstand ≥ 6 mm + Frässchlitz zur Niederspannung)



24 V Schiene



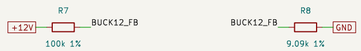
24 V → 5 V / 3 A (Stamp-P4 + 7"-Display)



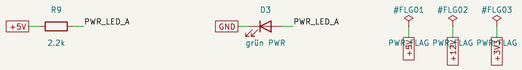
5 V → 3,3 V / 2 A (ESP32-S3, ADC/DAC, RS485, PHY) + USB-Versorgung für Werkbank



24 V → 12 V (Lüfter)



Anzeige & Power-Flags



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

Brunnensteuerung Wathlingen

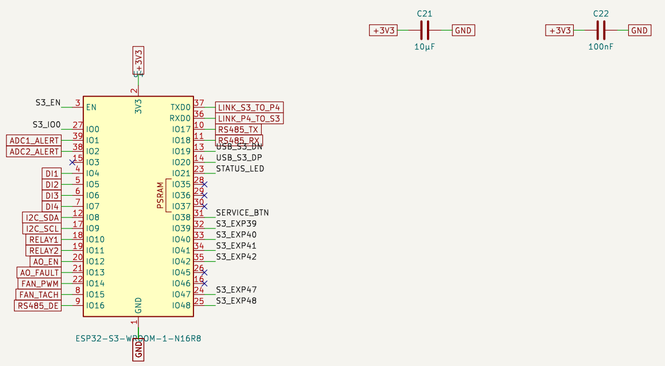
Sheet: /Stromversorgung/

File: power.kicad_sch

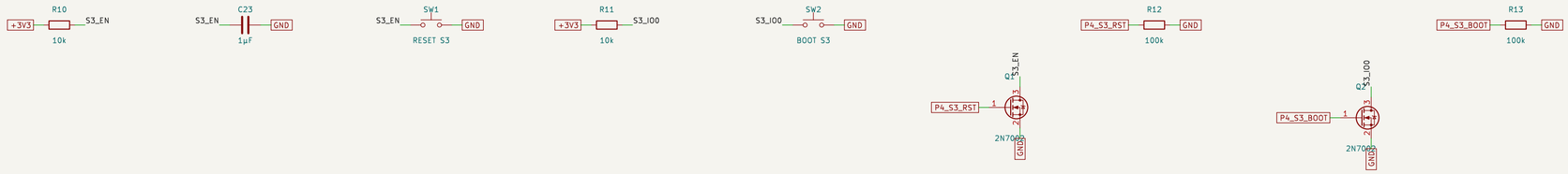
Title: Stromversorgung 230 V → 24 V / 12 V / 5 V / 3,3 V

Size: A2	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 2/7

ESP32-S3-WROOM-1-N16R8 (IO3/IO45/IO46 = Straps, IO35-37 = Octal-PSRAM → frei lassen)



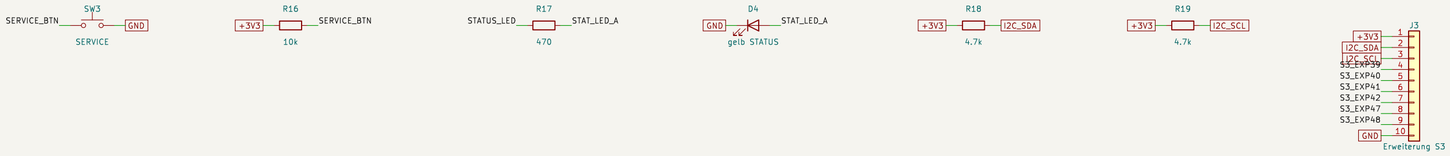
Reset / Boot (auch vom P4 aus steuerbar → P4 kann die Steuerung per OTA flashen)



USB-C (S3 flashen / Konsole; versorgt die Platine auf der Werkbank ohne 230 V)



Bedienung, LED, I²C-Pull-ups, Erweiterung



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

Brunnensteuerung Wathlingen

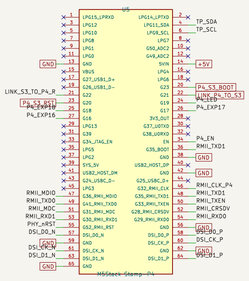
Sheet: /Steuerung_ESP32S3/

File: ctrl_s3.kicad_sch

Title: Steuerung: ESP32-S3-WROOM-1 (Regelung, I/O, V20, Sicherheit)

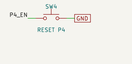
Size: A2	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 3/7

M5Stack Stamp-P4 (ESP32-P4, 16 MB Flash, 32 MB PSRAM) – eigener USB-C am Platinenrand zugänglich











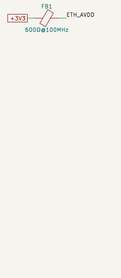


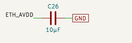


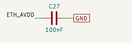




Ethernet-PHY LAN8720A (25-MHz-Quarz, REFCLKO 50 MHz → P4)

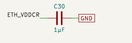




























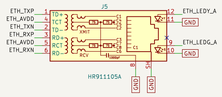








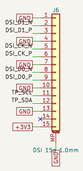








7"-Display: MIPI-DSI 2-Lane, FPC 15-pol. 1,0 mm (Raspberry-Pi-Standard) + 5-V-Versorgung













Befestigung



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

Brunnensteuerung Wathlingen

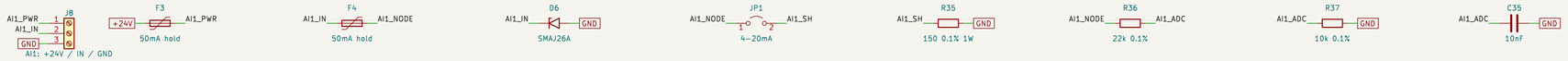
Sheet: /HMI_ESP32P4/

File: hmi_p4.kicad_sch

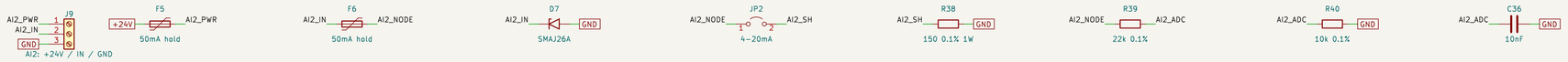
Title: App/HMI: M5Stack Stamp-P4, Ethernet (LAN8720A), 7"-DSI-Touch

Size: A1	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/7

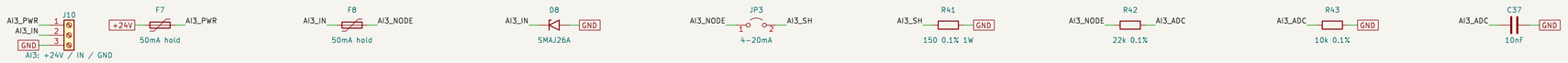
AI1 – Druck 0-10 V (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



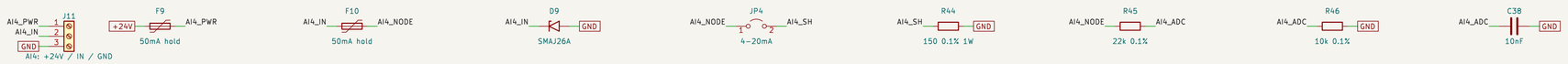
AI2 – Durchfluss 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



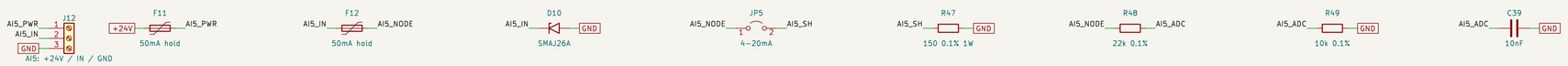
AI3 – Wassertemp 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



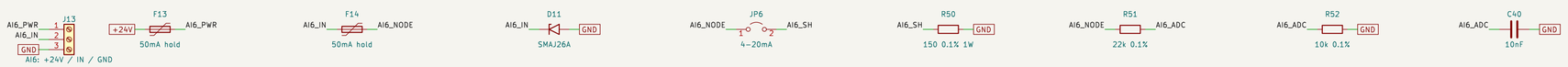
AI4 – Regentonne 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



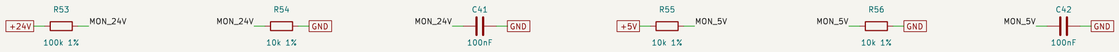
AI5 – Reserve (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



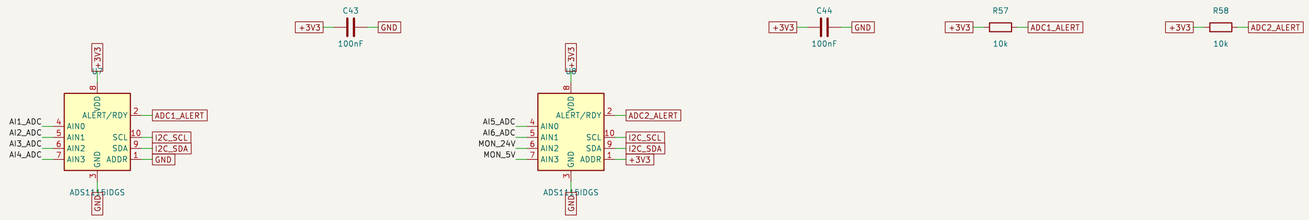
AI6 – Reserve (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



Spannungsüberwachung



ADC 1 (0x48: AI1-AI4) / ADC 2 (0x49: AI5, AI6, 24V, 5V)



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

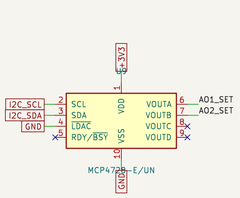
Brunnensteuerung Wathlingen

Sheet: /Analogeingänge/
File: analog_in.kicad_sch

Title: 6 × Universal-Analogeingang 0-10 V / 4-20 mA (Jumper)

Size: A2	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/7

DAC (I²C 0x60, interne Ref. 2,048 V) & Ausgangsfreigabe







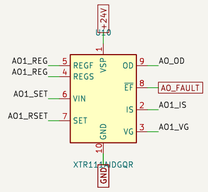


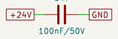






AO1 – Dosierpumpe   I = 10 · U_DAC / 1 kΩ (0…20,48 mA)







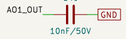










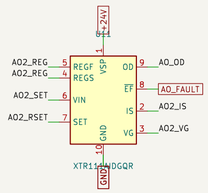








AO2 – Reserve   I = 10 · U_DAC / 1 kΩ (0…20,48 mA)













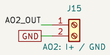












Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

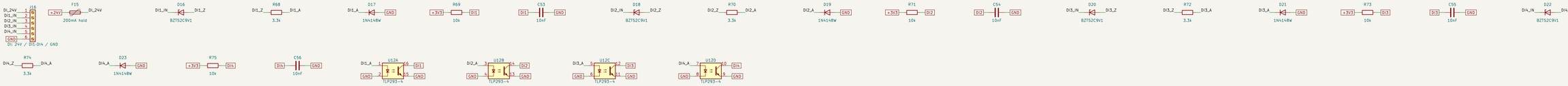
Brunnensteuerung Wathlingen

Sheet: /Analogausgänge/
File: analog_out.kicad_sch

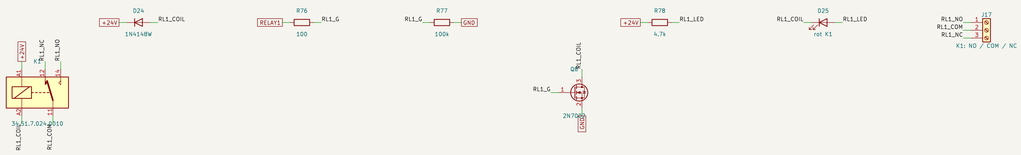
Title: 2 × Analogausgang 0/4-20 mA (Dosierpumpe + Reserve)

Size: A2	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

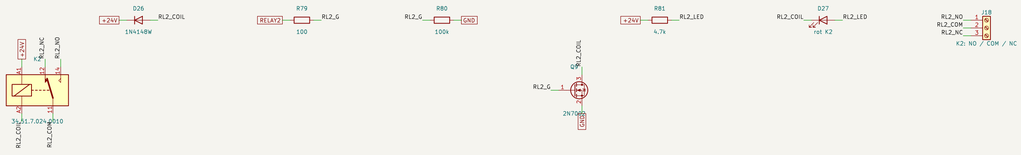
Digitaleingänge 24 V (Schaltschwelle ≈ 12 V, aktiv = GPIO LOW)



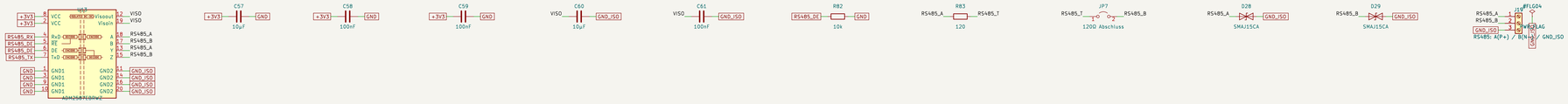
Relais K1 (230 V / 6 A, potentialfrei) – z. B. Freigabe Dosierpumpe



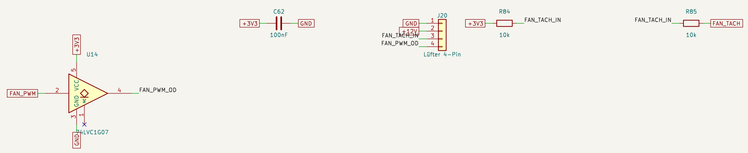
Relais K2 (230 V / 6 A, potentialfrei)



RS485 isoliert → V20 (Modbus RTU, Halbduplex, DE über RTS/GPIO17)



Gehäuselüfter 12 V, 4-Pin-PWM (GPIO18, 25 kHz) + Tacho (GPIO23)



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

Brunnensteuerung_Wathlingen

Sheet: /Digital_RS485_Luefter/

File: lo.kicad_sch

Title: Digitaleingänge, Relais, RS485 (V20), Lüfter

Size: A1	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11-dfsg-1build4		Id: 7/7